

Problem 11.9

Jennifer deposits 1000 into a bank account. The bank credits interest at a nominal annual rate of i compounded semi-annually for the first 7 years and a nominal annual rate of $2i$ convertible quarterly for all years thereafter. The accumulated amount in the account after 5 years is X . The accumulated amount in the account at the end of 10.5 years is 1,980. Calculate X .

Solution

For the first 7 years, the interest rate per half-year is $i/2$, and there are $2(7) = 14$ half-year periods. For the remaining $10.5 - 7 = 3.5$ years, the interest rate per quarter is

$$\frac{2i}{4} = \frac{i}{2},$$

and there are $4(3.5) = 14$ quarterly periods.

Thus, the accumulation factor per compounding period is the same in both intervals. The accumulated value at time 10.5 satisfies

$$1980 = 1000 \left(1 + \frac{i}{2}\right)^{14} \left(1 + \frac{i}{2}\right)^{14}.$$

Therefore,

$$1.98 = \left(1 + \frac{i}{2}\right)^{28}.$$

Taking the 28th root gives

$$1 + \frac{i}{2} = (1.98)^{1/28},$$

so

$$i = 2 \left((1.98)^{1/28} - 1 \right) \approx 0.04939268173.$$

At time 5, only the semi-annual rate has applied, for $2(5) = 10$ periods. Hence,

$$X = 1000 \left(1 + \frac{i}{2}\right)^{10} = 1000(1.98)^{10/28}.$$

Thus,

$$X \approx 1276.297309.$$

Therefore,

$$\boxed{X \approx 1276.30}.$$